CoolPAK™ S4428

Mineral oil free semi-synthetic

CoolPAK™ S4428 is an innovative, translucent semi-synthetic metalworking fluid free of mineral oil. The product is ideal for use in turning and milling of structural steels, alloy steels and titanium due to its well balanced lubrication and cooling characteristics. It can also be used for the machining and grinding of cast iron but is not suitable for yellow metal alloys. The product is low foaming even in soft waters but is also stable in waters with a high hardness.

Application Guidelines

- Performs well where traditional soluble oils may not cool sufficiently.
- Running at or above 7.0% offers the best sump life and corrosion inhibition on cast iron chips.
- Not recommended for use for yellow metals or very reactive metals such as magnesium.
- For additional product application information, including performance optimisation, please contact your Master Fluid Solutions' Authorised Distributor at https://www.masterfluids.com/eu/en/distributors/index.php, your District Sales Manager, or call our Tech Line at +49 211 41 72 8 -900.

Physical Properties Typical Data

Colour (Concentrate)	Light brown
Colour (Working Solution)	Translucent
Odour (Concentrate)	Bland
Form (Concentrate)	Liquid
Flash Point (Concentrate)	> 160°C (ASTM D93-08)
pH (Concentrate as Range)	9.4 - 9.6
pH (Typical Operating as Range)	8.5 - 9.4
Coolant Refractometer Factor	2.9
Digital Titration Factor	0.0307

Choose CoolPAK S4428:

- Formaldehyde releaser and mineral oil free
- Low foaming for modern high-pressure, high-volume applications
- Keeps machines very clean while leaving a soft fluid film for ease of cleaning and reduced maintenance
- Proven performer in titanium milling applications
- Greatly extends useful life without the need for tank side biocides or fungicides
- Provides superior corrosion inhibition on all ferrous metals
- Uses standard metalworking recycling and disposal techniques

Health and Safety

For further information, see the most recent SDS which is available directly from Master Fluid Solutions or from your Master Fluid Solutions' Authorised Distributor.

Packaging

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Recommended Concentrations

Light Duty	5.0% - 7.5%
Moderate Duty	7.5% - 9.5%
Heavy Duty	9.5% - 15.0%
Design Concentration Range	5.0% - 15.0%

Mixing Instructions

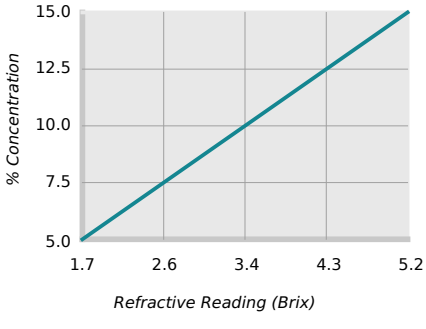
- Recommended usage concentration in water: 5.0% - 15.0%
- To help ensure the best possible working solution, add the required amount of concentrate to the required amount of water (never the reverse) and stir until uniformly mixed.
- Use mineral-free water to improve sump life and corrosion inhibition while reducing carryoff and concentrate usage.

Additional Information

- Use Master STAGES™ Whamex XT™ for a quick and thorough precleaning of your machine tool and coolant system.
- Consult Master Fluid Solutions before using on any metals or applications not specifically recommended.
- This product should not be mixed with other metalworking fluids or metalworking fluid additives, except as recommended by Master Fluid Solutions, as this may reduce overall performance, result in adverse health effects, or damage the machine tool and parts. If contamination occurs, please contact Master Fluid Solutions for recommended action.

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Concentration by % Brix



% Concentration = Refractive Reading x Refractive Factor
Coolant Refractometer Factor % Brix = 2.9

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